router. Hop limit of IPv6 is exactly time-to-live (TTL) in IPv4.

• IP address of the original source of the datagram is called source address. IP address of the destination of the datagram is called destination address. Each is of 128 bits. Three main groups of IP addresses in IPv6 are unicast, multicast and anycast.

Unicast address is for a particular host. A unicast packet is identified by its unique single address for a single interface network interface card (NIC) and is transmitted point-to-point.

A multicast address is for all hosts of a particular group to receive the datagram.

The anycast address is to address a number of interfaces on a single multicast address.

Unlike IPv4, IPv6 addresses do not have classes. Address space of IPv6 is sub-divided in various ways based on purpose of use. The notion for writing IPv6 address is each double byte of the 16-byte field is separated by a colon. Bytes are written in hexadecimal symbols. Thus, one example of the address could be 1234:ABCD:56EF:0079:0000:0000:9876:456C.

The proposed notation has the provision to skip leading zeros so that 0000 can be written as 0, or 0056 can be written as 56. It has also the provision of removing one 0, leaving the colons by the technique of double colon notation.

Address of the example given above therefore may be represented as:

1234:ABCD:56EF:79::9876:456C

The double colon notation is used at the beginning and the end of the address but only once. In transition period IPv4, the address would be converted to IPv6 address by pretending 12 bytes of 0s. Thus, IPv4 address 100.45.28.17 may be expressed as:

::100.45.28.17, as an equivalent IPv6 address.

It is found that several fields of IPv4 are no longer in IPv6. Notably among these are checksum, option and fragmentation fields. One of the primary goals of designing IPv6 was the fast processing of packets. Error check is done at upper layers, namely TCP/UDP in IPv6. With no option field, IPv6 is made of fixed size frame, reducing node processing delay. However, scope of extension of the header exists in IPv6. Scope of fragmentation is made a function of source only in IPv6, unlike at every router in IPv4.

Using extension headers

Hop-by-hop extension header. The payload length field of IPv6 defines the maximum payload size in bytes. Thus, maximum size of the payload is limited to 65,535 bytes. Many applications like those of multimedia services may require payloads in sizes larger than 65,535 bytes. Hop-by-hop extension header supports larger payloads, above 65,535 bytes. This extension header may increase the size of payload by 2^{19} bytes.

Header length of the extension header specifies the data in extension in bytes in multiples of 8 bytes. Therefore each extension header can accommodate about 2^8 bytes. The allowed maximum six extension headers limit may increase the size of the payload by six times of 2^8. Thus, maximum size of payload becomes about 2^{19} bytes.

There is also a provision for construction of jumbo packet formation with 32-bit field of payload header. Once specification of jumbo packet is adopted, payload size can be as large as 4,294,967,295 bytes (maximum).

Routing extension header. It is typically that of loose source and record routing (LSRR) option field of IPv4 and is used to provide addresses of routers that IPv6 packets shall follow to reach the destination node. This is useful for sending packets by the virtual circuit switching mode of store and forward technique. This makes IPv6 support time-sensitive services of large data.

Fragment extension header. It is used for the purpose of fragmentation of packets as in IPv4. Minimum IPv6 packet size is 1280 bytes. When the receiving node is on a network that has maximum transfer unit (MTU) and is less than 1280 bytes, the sender fragments the packets. By fragmentation, packets are made of the size that the network may carry.

Details of fragmentation are conveyed to the receiving node by fragment extension header. For example, when the packet is fragmented, each part of that packet is assigned the same identification number. Identification number is of 32 bits and the same is notified in variable extension header field. In IPv4, intermediate nodes may fragment data. In IPv6, only original sender or node can fragment data.

Authentication extension header. This is used to authenticate that the original packet sent by the sender is received perfectly by the receiver. When extension header is included, variable extension header fields will include the authenticated code like that of MD5 (message Digest 5) or digest by hash function of all headers. Note that the hop count field will change at each hop. Thus, digest is either excluded from the hop count field or its code is made 0 in authentication.

Encapsulating security payload (ESP) extension header. This is used to provide data security to the payload only by means of encryption/decryption technique, ES (Data Encryption Standard). When this header is included, the sender provides security information in variable extension header fields. The receiver using the information in variable extension header fields decrypts the payload.

Internet Control Message Protocol

IPv6 has dropped the fragmentation and reassembly feature at intermediate routers. Data, if required, is fragmented for packetisation at the source only. Reassembling is done at destination only. When an IPv6 packet received by any intermediate router is too large to be carried forward on the outgoing link, the router simply drops the packet and then sends an ICMP error message to the sender. Error message so sent reads like Packet Too Big. The sender on receiving the ICMP error message Packet Too Big fragments the packet in a smaller size and then transmits fragmented packets, each as one unit.

ICMP for IPv4 is used by hosts, nodes, routers and gateways to communicate network layer information to each other. ICMP information is carried as IP payload like TCP or UDP information. ICMP messages are basically used for error reporting, among